# David Hyunyoo Jang

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## Education

**Seoul National University** **Mar 2022 – Feb 2028 (Expected)**
*PharmD* *Seoul, South Korea*

**Korea University** **Mar 2019 – Aug 2021**
*Coursework in Materials Science and Engineering (no degree)* *Seoul, South Korea*

## Research Experience

**AI Protein Design Lab**, Korea Advanced Institute of Science and Technology (KAIST) **Jun 2025 – Aug 2025**
*Undergraduate Researcher (advisor: Gyu Rie Lee)* *Daejeon, South Korea*

**Collaborative small-molecule sensor binder design campaigns**
- Developed, standardized, and released a reproducible end-to-end binder design workflow during phenytoin and carbamazepine sensor-design campaigns, integrating diffusion-based backbone generation, LigandMPNN, Rosetta scoring, AF3/Boltz-2 refolding, PLACER, TMalign, and RMSD-based triage with parquet-based metric lineage and SLURM-scalable execution. [code]
- Developed metric-guided LigandMPNN–FastRelax recycling to diversify and optimize candidate pools beyond sequence-only sampling across phenytoin/carbamazepine campaigns; in the carbamazepine campaign, expanded screened candidates (*21→275*) and improved the best Rosetta interface $\Delta\Delta G$ score (*–25.4→–31.6*). [code]

**RFdiffusion2 atom-wise RASA conditioning for ligand orientation control**
- Implemented inference-time atom-wise RASA conditioning for RFdiffusion2 to control buried/exposed ligand atoms and improve ligand-orientation control while preserving compatibility with global RASA inputs. [PR]

**Laboratory of Computational Drug Discovery**, Seoul National University **Jul 2023 – Aug 2025**
*Undergraduate Researcher (advisor: Juyong Lee)* *Seoul, South Korea*

**PREpiBind — pMHC-II binding: representation benchmark & web server**
- Independently built PREpiBind, a joint-attention framework benchmarking 10 protein representations under a fixed architecture — BLOSUM62, structure-prediction embeddings (AlphaFold3/Chai-1/Boltz-1, extracted by modifying inference code), and protein language models (PLMs: ESMC/ESM3) — for pMHC class II binding on IEDB qualitative, MS, and IC50 datasets; PLMs generally led pooled performance, exceeding NetMHCIIpan-4.3, MixMHC2pred-2.0, and DeepNeo on qualitative and MS benchmarks (NetMHCIIpan-4.3 led on thresholded IC50).
- Built leakage-controlled splits and stress-tested generalization under leave-one-molecule-out and mouse (H2) held-out validation, where PLM gains narrowed; re-implemented DeepNeo in PyTorch (ROC-AUC 0.894; published 0.882).
- Showed that ESMC 300M matches larger PLMs (ESMC 600M, ESM3 Small/Medium) within seed variation (ROC-AUC 0.919–0.921); released code (MIT license), datasets, and trained checkpoints. [preprint; code]
- Built a public web server (precomputed embeddings for 7,282 IPD-IMGT/HLA and mouse H2 chains; zero-shot) with a custom-sequence mode for non-human alleles and Colab/CLI demos; up to ∼69× faster than command-line tools. [server]

**LLMs on the Korean Pharmacist Licensing Exam**
- Designed and built an end-to-end benchmark of 27 LLMs — 7 proprietary and 20 open-source (4–104B) — across six KPLE exam years (2019–2024) in Korean and English, covering question extraction (text + vision-OCR), LLM-based translation, local/API inference, and official scoring with human-percentile normalization.
- 7 models passed all years in both languages — the top reaching the top 12% of human examinees — under a logarithmic size–performance trend; developed a multi-pronged data-contamination analysis arguing against pretraining leakage as the dominant performance driver. [preprint; code]

**CB1 receptor small-molecule virtual screening**
- Reproduced an end-to-end structure-based virtual screen of the Enamine Hit Locator Library (460K compounds) against the CB1 receptor on an 8-GPU cluster (∼12 h) — self-compiling AutoDock-GPU from source for multi-GPU execution (the prebuilt binary used 1 of 8 GPUs) — and automating batch docking, result parsing, and ranking.

**Structure & Knowledge Injection into Machine Learning Lab**, Seoul National University **May 2024 – Sep 2024**
*Undergraduate Researcher (advisor: Jay-Yoon Lee)* *Seoul, South Korea*

**Semi-supervised data augmentation for RoseTTAFold2NA** *collaboration with Minkyung Baek's lab*
- Implemented the training pipeline for a semi-supervised joint-loss objective that augments RF2NA with unlabeled self-distilled structures — built the mixed labeled/unlabeled data loader, per-batch loss routing, and a distributed sampler with tunable unlabeled fraction, consolidating scattered prototype scripts into one parameterized training entry point.
- Designed and ran a controlled low-label study — PDB-only vs distilled-augmentation vs joint-loss semi-supervision across 10/30/50% label-density regimes — to test whether unlabeled data improves RF2NA learning under sparse supervision.

**Systems Pharmacology Lab**, Seoul National University **Jun 2023 – Jul 2023**
*Undergraduate Internship (advisor: Minsoo Noh)* *Seoul, South Korea*
- Performed HaCaT cell culture and $PGE_2$ competitive ELISA to assess anti-inflammatory effects of tryptophol; conducted dose–response analysis and benchmarking against NSAIDs.

## Publications & Preprints

**[1]** PREpiBind: Protein Representation-integrated Epitope–MHC Class II Binding Prediction
**David Hyunyoo Jang**, Dongwoo Kim, Untaek Hwang, Byungho Park, Yoonjoo Choi, Juyong Lee
Submitted; preprint: *bioRxiv* (2026), doi:10.64898/2026.09.15.751749

**[2]** Proprietary and Open-Source Large Language Models on the Korean Pharmacist Licensing Examination: A Comparative Benchmarking Study
**David Hyunyoo Jang**, Juyong Lee
Under review; preprint: *medRxiv* (2025), doi:10.1101/2025.04.15.25325584

## Selected Research Projects

**foldspace** **2026 – present**
*Self-initiated; toward an ESMFold2-based design model*

**design_cycle** **2026 – present**
*Self-initiated; follow-up to LigandMPNN–FastRelax recycling*

## Research Software & Open-Source Contributions

- **ESMFold2–AtomWorks** — integrated ESMFold2 with AtomWorks data structures and an optional Foundry training path; exact feature parity on reference fixtures and exact output parity under deterministic GPU kernels. [code]
- **RosettaCommons Foundry** — merged LigandMPNN inference improvements: per-design and ligand-interface confidence reporting, and parser overrides for AtomArray inputs. [#306; #389]
- **Upstream fixes** — AtomWorks insertion-code residue parsing, proposed and incorporated upstream in #116 [#107]; ESMFold2 multi-component CCD ligand output, the same change now present upstream [#416].
- **patinae** (Rust/WGPU molecular viewer) — fixed renderer failures on low-limit GPUs and Windows. *Merged* [#28; #29]

## Presentations

**[1]** PREpiBind: Protein Representation-integrated Epitope–MHC Class II Binding Prediction
KIDDS 2025 Summer Workshop (Korean In silico bioDesign and Discovery Society), Poster

**[2]** DeepXeno: Structure-Informed Prediction of Xenotransplantation Immunogenicity *(model later released as PREpiBind)*
2024 SNU Student-Directed Education Undergraduate Research Program, Talk (Outstanding Research Presentation)

## Awards & Funding

**Research Excellence Award**, 2024 SNU Student-Directed Education Undergraduate Research Program **2025**
*President of Seoul National University*

- Awarded to two projects per fields (ten total) for excellence in research execution and outcomes.

**2024 SNU Student-Directed Education Undergraduate Research Program** **2024**
*Faculty of Liberal Education, Seoul National University*

- Selected as one of 36 projects to receive funding for undergraduate research. *(KRW 7M)*

## Teaching & Leadership

**SNU Alpha** | Deep Learning Study Club **Spring 2022 – Winter 2024**
*Executive Member, Co-Founder* *College of Pharmacy, Seoul National University*

- Co-founded and helped organize a peer-led deep learning study club for pharmacy students.
- Led weekly study sessions on Natural Language Processing (Fall 2023) and PyTorch (Spring 2024), teaching deep learning concepts and applications in computational drug discovery.

**Military Service** **Aug 2025 – Feb 2027 (Expected)**
*Team Leader, Sergeant* *III Corps, Republic of Korea Army*

- Led a 10-person air-defense team, coordinating personnel and overseeing the execution of assigned duties.
- Operated and maintained air-defense radar systems in a frontline sector along the Korean DMZ.

## Clinical Pharmacy Training

**Seoul National University Hospital** **Dec 2024 – Feb 2025**

*Hospital Pharmacy Practicum* *Seoul, South Korea*

- Completed rotations across 12 specialty departments, participating in medication dispensing, drug therapy case reviews, and patient counseling under pharmacist supervision.

**E-Blue Pharmacy** **Feb 2025 – Mar 2025**

*Community Pharmacy Practicum* *Suwon, South Korea*

- Reviewed prescriptions for potential medication-related problems and participated in medication therapy reviews and counseling on prescription and over-the-counter medications under pharmacist supervision.

## Technical Skills

**Programming & Data**: Python, PyTorch, Numpy, Bash, pandas, matplotlib, Streamlit, Nginx
**Modeling & Docking**: RDKit, PyMOL, Chimera, AutoDock, PyRosetta, RFdiffusion (AA), LigandMPNN, AlphaFold3, ESMFold2/ESMC, AtomWorks/Foundry, TMol, Biotite
**Systems & DevOps**: Linux, Git, Slurm, conda/mamba, HuggingFace
**Experimental Techniques**: cell culture (HaCaT), buffer/reagent prep, competitive ELISA, serial dilutions, plate reader